

Python Basics - Day 3 Notes

Topics Covered: Lists and Tuples

1. Lists:

- Ordered, mutable collection (can be changed)
- Syntax: `list_name = [item1, item2, item3]`
- Access: `list[index]`, slicing with `list[start:end]`
- Common Methods: `append()`, `insert()`, `remove()`, `pop()`, `sort()`, `reverse()`

Example:

```
fruits = ['apple', 'banana', 'mango']
fruits.append('grape')
fruits[1] = 'orange'
```

2. Tuples:

- Ordered, immutable collection (cannot be changed)
- Syntax: `tuple_name = (item1, item2, item3)`
- Access: `tuple[index]`, slicing works too

Example:

```
person = ('Harshith', 34, 'Bangalore')
print(person[0])
# person[1] = 35 ■ Error
```

3. Lists vs Tuples:

- Lists: mutable, slower, use when data changes
- Tuples: immutable, faster, use for fixed data

4. Conversion:

- `list()` converts tuple → list
- `tuple()` converts list → tuple

Example:

```
t = ('a', 'b')
l = list(t); l.append('c'); t = tuple(l)
```

Practice Examples:

- Sum, max, min of list [1,2,3,4,5]
- Append 'mango', remove 'apple' from fruit list
- Tuple person info and conversion to list